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Workshop -1

This workshop will test your knowledge of basic C concepts such as variable types, conditions, and loops.

1. Create a program to classify a student's grade into 3rd, 2:2, 2:1, and first from taking in a percentage number (integers). 3rd is <50, 2:2 is <60, 2:1 is <70 and first is >=70.

Ans :

Source Code:

```c

#include<stdio.h>
void main(){
  int percentage;
  char choice;
  while(1){
  printf("Enter your percentage :");
  scanf("%d",&percentage);

  if(percentage>=70 && percentage<=100){
    printf("You are first.");
   }else if(percentage>=60 && percentage<70){
    printf("You are 2:1.");
   }else if(percentage>=50 && percentage<60){
    printf("You are 2:2.");
   }else if(percentage>=0 && percentage<50){
    printf("You are 3rd.");
   }else{
    printf("Invalid input");
   }
   printf("\nPress n to end.");
   scanf(" %c",&choice);
   if(choice=='n'){
     break;
   }
  }
  printf("\n");

}
```

Output:

```
Enter your percentage :90
You are first.
Press n to end.a
Enter your percentage :65
You are 2:1.
Press n to end.h
Enter your percentage :55
You are 2:2.
Press n to end.y
Enter your percentage :22
You are 3rd.
Press n to end.f
Enter your percentage :-65
Invalid input
Press n to end.g
Enter your percentage :100000
Invalid input
Press n to end.n
```

2. Build on question one by using a list of percentages (integers) and classify all grades.

Ans:

Source Code:

```c
#include<stdio.h>
void main(){
  int n;
  printf("Enter how many grades:");
  scanf("%d",&n);

  int percentages[n];
  for(int i=0;i<n;i++){
  printf("Enter percentage of student number : %d :",i+1);
  scanf("%d",&percentages[i]);
  }

  for(int i=0;i<n;i++){
  if(percentages[i]>=70 && percentages[i]<=100){
    printf("Percentage classification of student number %d: First.\n",i+1);
   }else if(percentages[i]>=60 && percentages[i]<70){
    printf("Percentage classification of student number %d: 2:1.\n",i+1);
   }else if(percentages[i]>=50 && percentages[i]<60){
    printf("Percentage classification of student number %d: 2:2.\n",i+1);
   }else if(percentages[i]>=0 && percentages[i]<50){
    printf("Percentage classification of student number %d: 3rd.\n",i+1);
   }else{
    printf("Invalid input\n");
   }
  }
  printf("\n");
  }
```

Output:

```
Enter how many grades:5
Enter percentage of student number : 1 :90
Enter percentage of student number : 2 :60
Enter percentage of student number : 3 :51
Enter percentage of student number : 4 :9000
Enter percentage of student number : 5 :22
Percentage classification of student number 1: First.
Percentage classification of student number 2: 2:1.
Percentage classification of student number 3: 2:2.
Invalid input
Percentage classification of student number 5: 3rd.
```

3. Create a program that uses a while loop to print the first 50 triangular numbers. This question requires two bits of research: one is the formula to detect a triangular number (look online), and the other is to figure out when to stop. HINT – every time you print a triangular number, increment the loop variable!

Ans:

Source Code:

```c
#include<stdio.h>
void main(){
 printf("First 50 triangular numbers : \n");
 for(int i=1;i<=50;i++){
  int tn=(i*(i+1))/2;
  printf("%d\n",tn);
 }
 printf("\n");
}
```

Output:

```
First 50 triangular numbers :
1
3
6
10
15
21
28
36
45
55
66
78
91
105
120
136
153
171
190
210
231
253
276
300
325
351
378
406
435
465
496
528
561
595
630
666
703
741
780
820
861
903
946
990
1035
1081
1128
1176
1225
1275
```

4. Declare an array of at least 11 integers; the numbers can be random, so you can choose them. Create a program to loop through that list and print the:
    a. Smallest value
    b. Largest value
    c. Average (use float or double)

Ans:

Source Code:

```

#include<stdio.h>
void main(){
 int numbers[11],smallest,largest,sum=0;
 float average;

 printf("Enter 11 integers.\n");
 for(int i=0;i<=10;i++){
  printf("Number %d : ",i+1);
  scanf("%d",&numbers[i]);
  printf("\n");
 }
 smallest=numbers[0];
 largest=numbers[0];
 for(int i=1;i<=10;i++){
  if(numbers[i]>largest){
  largest=numbers[i];
  }
  if(numbers[i]<smallest){
  smallest=numbers[i];
  }
  sum=sum+numbers[i];
 }

 average=sum/11;
 printf("Smallest value : %d",smallest);
 printf(" \nLargest value : %d",largest);
 printf(" \nAverage value : %f",average);

 printf("\n");
}
```

Output:

```
Enter 11 integers.
Number 1 : 3

Number 2 : 4

Number 3 : 457

Number 4 : 14

Number 5 : 856436

Number 6 : 135436

Number 7 : 12643

Number 8 : 2142325

Number 9 : 626635324

Number 10 : 34532

Number 11 : 14354

Smallest value : 3
Largest value : 626635324
Average value : 57257412.000000
```

5. Create a simple login system with a fixed pin. Create an integer called pin and set this to any 4-digit number. Then ask the user to type in a value; if this does not match, the user is given another 5 attempts. Your program should also log how many attempts it took to get the user to type in the correct pin.

Ans:

Source Code:

```c
#include<stdio.h>
#include<stdbool.h>

void main(){
 int fixed_pin=2006;
 int pin;
 bool logged_in=false;

 for(int i=4;i>=0;i--){
 printf("Enter your PIN : ");
 scanf("%d",&pin);
 if(pin==fixed_pin){
  printf("Successfully logged in in %d attempts\n",5-i);
  logged_in=true;
  break;
 }else{
  printf("Incorrect PIN. You have %d attempts left.\n",i);
 }
}

if(!logged_in){
 printf("You have been locked out for too many incorrect PIN attempts.");
 }
 printf("\n");
 }
```

Output:

```
Enter your PIN : 2345
Incorrect PIN. You have 4 attempts left.
Enter your PIN : 2006
Successfully logged in in 2 attempts
```

```
Enter your PIN : 3542
Incorrect PIN. You have 4 attempts left.
Enter your PIN : 252
Incorrect PIN. You have 3 attempts left.
Enter your PIN : 2141
Incorrect PIN. You have 2 attempts left.
Enter your PIN : 5356
Incorrect PIN. You have 1 attempts left.
Enter your PIN : 2341
Incorrect PIN. You have 0 attempts left.
You have been locked out for too many incorrect PIN attempts.
```

6. Ask the user for “n” and print the first n Fibonacci numbers
Ans:

Source Code:

```c
#include<stdio.h>
void main(){
 int n,a=0,b=1,c;
 printf("Enter how many numbers : ");
 scanf("%d",&n);
 printf("First %d Fibonacci numbers : \n",n);
 printf("%d \t %d",a,b);
 for(int i=2;i<n;i++){
   c=a+b;
   printf("\t%d",c);
   a=b;
   b=c;
 }
 printf("\n");
}
```

Output:

```
Enter how many numbers : 10
First 10 Fibonacci numbers :
0        1        1        2        3        5        8        13       21       34
```

7. Create a program to ask a user for a number, then check if this is a prime number.
    a. Extend this program by continuingly asking the user for a number until they type in a prime number to end the script

Ans:

Source Code:

```c
#include <stdio.h>
#include <stdbool.h>

void main(){
 int num;

 while(1){
 bool is_prime=true;

 printf("Enter a prime number to end this program:");
 scanf("%d",&num);
 if(num<=1){
   printf("%d is not a prime number.\n",num);
   continue;
 }
 for(int i=2;i*i<=num;i++){
  if(num%i==0){
   printf("%d is not a prime number.\n",num);
   is_prime=false;
   break;
 }
}
 if(is_prime){
  printf("%d is a prime number.\n",num);
  break;
 }
}
}
```

Output:

```
Enter a prime number to end this program:34
34 is not a prime number.
Enter a prime number to end this program:66
66 is not a prime number.
Enter a prime number to end this program:123456789
123456789 is not a prime number.
Enter a prime number to end this program:1
1 is not a prime number.
Enter a prime number to end this program:-5
-5 is not a prime number.
Enter a prime number to end this program:7
7 is a prime number.
```

8. Create a vending machine algorithm in C. For this program, you will need to start with 3 variables: one is the input (money put into the machine), the cost (product/s total), and the change to be dispensed. Your program should print out the coin value and coin amount for the change. For example, if my input was 250 pence (£2.50), the cost is 167 pence (£1.97),and the change is 53p.

Ans:

Source Code:

```c
#include<stdio.h>
void main(){
 int item_number,input,change,cost;
 while(1){
  cost =0;
  printf("What do you want : \n1.Chips \n2.Kit Kat \n3.Oreos \n4.Coke \n5.Water\n");
  scanf("%d",&item_number);
  switch(item_number){
   case 1:
    cost=167;
    break;
   case 2:
    cost=241;
    break;
   case 3:
    cost=80;
    break;
   case 4:
    cost=599;
    break;
   case 5:
    cost=99;
    break;
   default:
    printf("Invalid item number.\n");
  }
  if(cost==0){
   continue;
  }
  printf("Price : %d\n",cost);
  break;
  }

  while(1){
   printf("Insert money : ");
   scanf("%d",&input);
   if(input<cost){
    printf("Insufficient money\n");
    continue;
   }
   break;
  }
  change=input-cost;
  printf("Your change : %d pence.\n",change);
 }
```

Output:

```
What do you want :
1.Chips
2.Kit Kat
3.Oreos
4.Coke
5.Water
4
Price : 599
Insert money : 200
Insufficient money
Insert money : 1000
Your change : 401 pence.
```